

Physics A (H156, H556)

Exploring-Magnetism-MCQ+MS KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

Candidate forename		Candidate surname	
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Centre number						Candidate number				
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INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **24**.
- The total number of marks may take into account some 'either/or' question choices.

- 1 At the Earth's equator the magnetic flux density B is approximately $25 \mu\text{T}$.

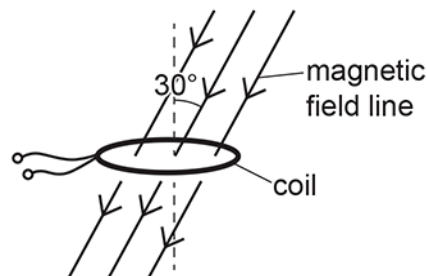
What is the magnitude of the force on an electron with velocity $v = 100 \text{ km s}^{-1}$ as it is moving perpendicular to the Earth's magnetic field at the equator?

- A $4.0 \times 10^{-25} \text{ N}$
- B $4.0 \times 10^{-22} \text{ N}$
- C $4.0 \times 10^{-19} \text{ N}$
- D $4.0 \times 10^{-16} \text{ N}$

Your answer

[1]

- 2 A flat coil has 200 turns and a cross-sectional area of $1.20 \times 10^{-4} \text{ m}^2$.



The coil is placed horizontally in a uniform magnetic field. The magnetic flux density is 0.050 T . The magnetic field is at an angle of 30.0° to the vertical.

What is the magnetic flux linkage for this coil?

- A $3.00 \times 10^{-6} \text{ Wb}$
- B $5.20 \times 10^{-6} \text{ Wb}$
- C $6.00 \times 10^{-4} \text{ Wb}$
- D $1.04 \times 10^{-3} \text{ Wb}$

Your answer

[1]

3 Faraday's law of electromagnetic induction is written below with two terms missing.

The induced in a circuit is directly proportional to the rate of change of magnetic flux

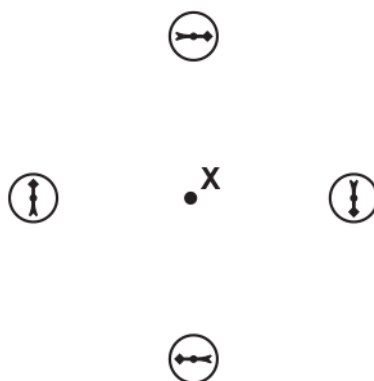
What are the **two** missing terms?

- A current, density
- B current, linkage
- C electromotive force, density
- D electromotive force, linkage

Your answer

[1]

4 The diagram shows four magnetic compasses placed at the same distance from point X.



Which of the following is most likely to be at point X?

- A permanent magnet
- B current-carrying solenoid
- C current-carrying flat coil
- D straight current-carrying wire

Your answer

[1]

5 Which law indicates that charge is conserved?

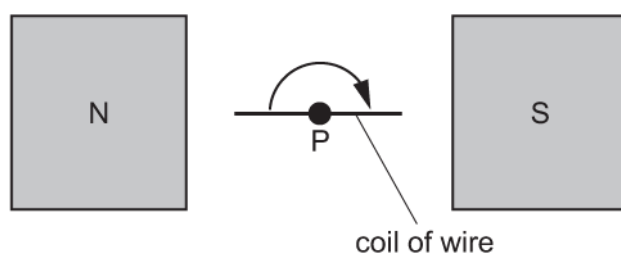
- A Lenz's law
- B Coulomb's law
- C Kirchhoff's first law
- D Faraday's law of electromagnetic induction

Your answer

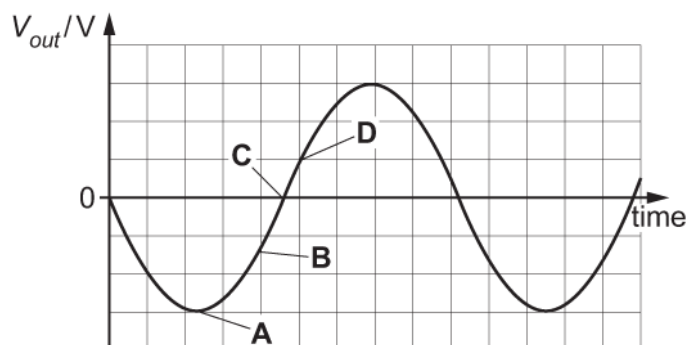
[1]

6 The diagram shows a coil of wire rotating between two permanent magnets in a model generator.

The coil is rotating clockwise about point P at constant angular velocity.



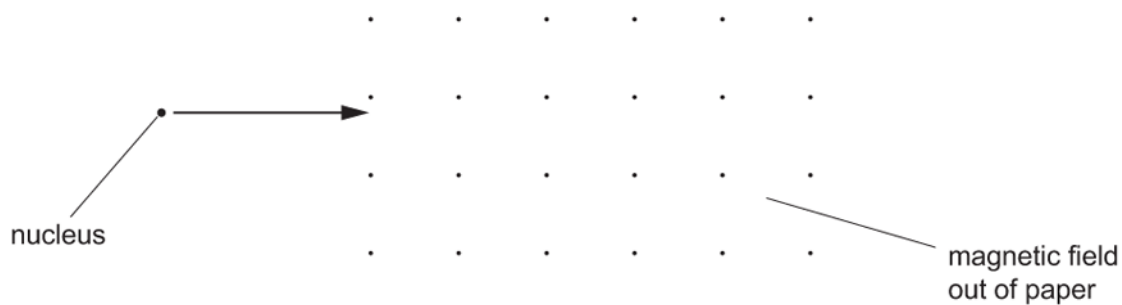
Which letter represents the output of the generator at the instant in the diagram?



Your answer

[1]

- 7 The diagram shows the path of a nucleus entering a magnetic field.



In which direction does the force on the nucleus act as it enters the magnetic field?

- A down the page
- B into the page
- C out of the page
- D up the page

Your answer

[1]

- 8 A particle is moving at right angles to a uniform magnetic field of flux density B . The particle has mass m , charge q and moves in a circular arc of radius r in the region of the magnetic field.

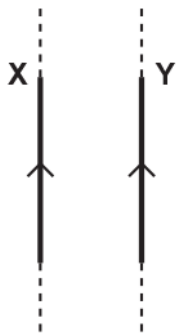
What quantities are required to determine the momentum of this particle?

- A B , q and r
- B B , q and m
- C B , q , r and m
- D q , r and m

Your answer

[1]

- 9 The diagram below shows two long current-carrying conductors X and Y.



The conductors are parallel to each other.

Y experiences a force because it is in the magnetic field of X.

Which row gives the correct direction of the magnetic field at Y due to X, and the direction of the force experienced by Y due to this field?

	Direction of magnetic field	Direction of force
A	Down into the plane of paper	To the right
B	Up from the plane of paper	To the right
C	Down into the plane of paper	To the left
D	Up from the plane of paper	To the left

Your answer

[1]

- 10 A coil with 500 turns is placed in a uniform magnetic field.
The average cross-sectional area of the coil is $3.0 \times 10^{-4} \text{ m}^2$.
The magnetic flux through the plane of the coil is reduced from $1.8 \times 10^{-4} \text{ Wb}$ to zero in a time t .
The average electromotive force (e.m.f.) induced across the ends of the coil is 0.75 V.

What is the value of t ?

- A $3.6 \times 10^{-5} \text{ s}$
B $2.4 \times 10^{-4} \text{ s}$
C 0.12 s
D 8.3 s

Your answer

[1]

- 11 The number of turns on the coils of four ideal iron-cored transformers A, B, C and D are shown in the table below.

Transformer	Number of turns on the secondary coil	Number of turns on the primary coil
A	100	100
B	50	200
C	200	50
D	500	100

Each transformer is connected in turn to an alternating 240 V supply.

Which transformer will give the largest output current?

Your answer

[1]

- 12 A beam of charged particles is not deflected when it passes through a region where both electric and magnetic fields are present.

Which statement is **not** correct?

- A All the particles have the same speed.
- B The resultant force on each particle is zero.
- C The magnetic force is equal to the electric force on each particle.
- D The magnetic field and the electric field are in the same direction.

Your answer

[1]

- 13 A current-carrying solenoid has N turns and radius r . The magnetic flux density within the core of the solenoid is B .

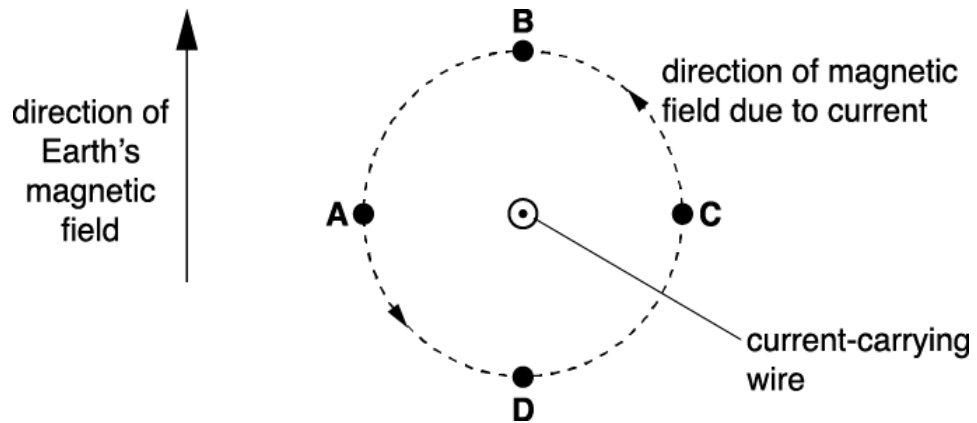
What is the magnetic flux linkage for this solenoid?

- A NB
- B $\pi r^2 B$
- C $2\pi rBN$
- D $\pi r^2 BN$

Your answer

[1]

- 14 The diagram below shows a current-carrying wire coming out from the plane of the paper. The current in the wire produces a magnetic field in an **anticlockwise** direction around the wire.



The direction of the Earth's magnetic field is also shown.

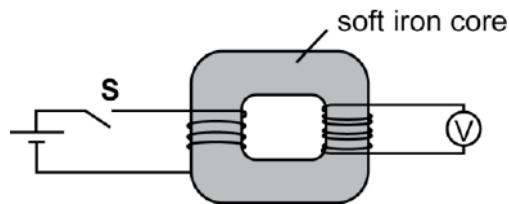
The Earth's magnetic field interacts with the magnetic field of the current-carrying wire.

At which point **A**, **B**, **C** or **D** is the resultant magnetic field strength a **minimum**?

Your answer

[1]

15 The diagram below shows a transformer.



The primary coil is connected to a switch **S** and a cell. The secondary coil is connected to a voltmeter. The switch is then **closed**.

Which statement is correct?

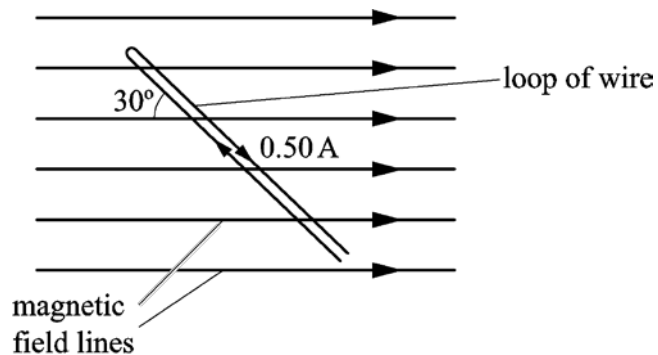
The voltmeter reading....

- A does not change.
- B increases and then stays constant.
- C increases and then decreases to zero.
- D increases and then changes direction.

Your answer

[1]

- 16 A rigid loop of insulated wire is placed in a uniform magnetic field of flux density 80 mT. The current in this loop is 0.50 A and the angle between the wire and the direction of the magnetic field is 30°.



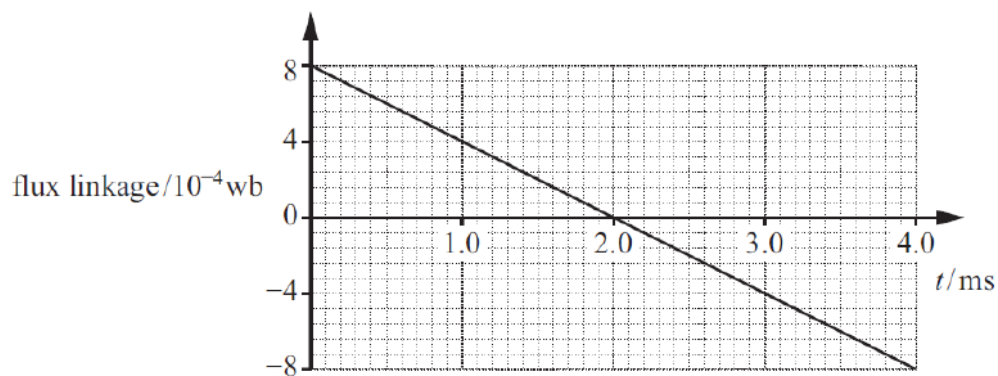
What is the magnitude of the force experienced by a 1.0 cm section of the loop?

- A. 0 N
- B. 2.0×10^{-4} N
- C. 3.5×10^{-4} N
- D. 4.0×10^{-4} N

Your answer

[1]

- 17 A coil with three turns of wire is used in an experiment.
The graph shows the variation of magnetic flux linkage with time t for this coil.



What is the e.m.f. induced across the ends of the coil?

- A. 0 V
- B. 0.20 V
- C. 0.40 V
- D. 1.2 V

Your answer

[1]

- 18 A proton with kinetic energy 1.6×10^{-17} J is travelling at right angles to a magnetic field. The magnetic flux density is 0.050 T.

What is the magnitude of the force on the proton?

- A 8.0×10^{-19} N
- B 1.1×10^{-15} N
- C 2.8×10^{-6} N
- D 6900 N

Your answer

[1]

- 19 A step-down transformer has an input potential difference of 200 V. There are 250 turns on the primary coil and 50 turns on the secondary coil. The secondary coil is connected to a 1.0 k Ω resistor.

What is the current through the resistor?

- A 2×10^{-4} A
- B 0.04 A
- C 1 A
- D 40 A

Your answer

[1]

- 20 Which statement is Faraday's law?

- A The direction of electric current induced by a changing magnetic field is such that the magnetic field created by the induced current opposes changes in the initial magnetic field.
- B The magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitudes of charges and inversely proportional to the square of the separation.
- C The magnitude of induced EMF is proportional to the rate of change of the magnetic flux linkage.
- D The total energy of an isolated system remains constant.

Your answer

[1]

- 21 A student is doing an experiment on the magnetic force experienced by a current-carrying wire in a uniform magnetic field. The magnetic flux density B can be varied.

For a particular flux density, the current in the wire is 2.0 A. The length of the wire in the field is 0.12 m. The angle between the current and the magnetic field is 30°. The force experienced by the wire is 7.7×10^{-2} N.

The student calculates B and records the results in a table.

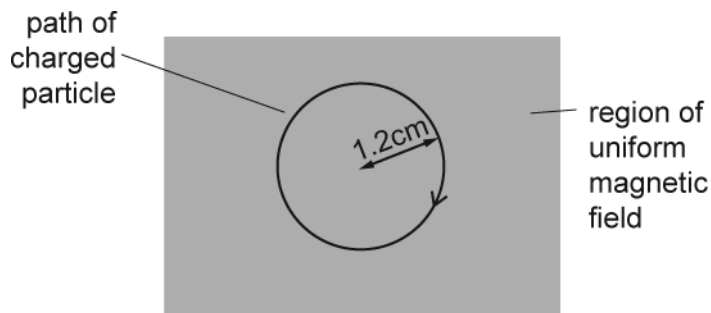
Which row shows the correct table heading for B and the correct value for B ?

	Table heading for B	Value for B
A	B / T	0.37
B	B / T	0.64
C	B / Wb	0.37
D	B / Wb	0.64

Your answer

[1]

- 22 A charged particle moves in a circular path of radius 1.2 cm in a uniform magnetic field.



The direction of the magnetic field is perpendicular to the plane of the paper.

The particle has mass m , charge $+Q$ and speed v .

Another particle of mass $3m$, charge $+2Q$ and speed v moves in a circular path of radius R in the same magnetic field.

What is the value of R ?

- A 0.8 cm
- B 1.2 cm
- C 1.8 cm
- D 7.2 cm

Your answer

[1]

- 23 An electron moves in a circle of radius 2.0 cm in a uniform magnetic field of flux density 170 mT.

What is the momentum of this electron?

- A $3.4 \times 10^{-3} \text{ kg m s}^{-1}$
- B $5.4 \times 10^{-17} \text{ kg m s}^{-1}$
- C $1.4 \times 10^{-18} \text{ kg m s}^{-1}$
- D $5.4 \times 10^{-22} \text{ kg m s}^{-1}$

Your answer

[1]

- 24 A charged particle travelling with speed v describes a circular path of radius R in a plane perpendicular to a uniform magnetic field. The orbital period of this particle is T .

The same particle now travels with speed $2v$ in a circular path in the same plane as before.

What is the orbital period of the particle now?

- A $0.25T$
- B $0.5T$
- C T
- D $2T$

Your answer

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1			C	1	<u>Examiner's Comments</u> Most candidates were able to calculate the force, using $F = BQv$. The powers of ten in the various quantities appeared to have caused few problems. This was the multiple choice question that had the most 'no responses'.
			Total	1	
2			D	1	<u>Examiner's Comments</u> The correct response is D. Electromagnetism is another challenging set of concepts, resulting in a relatively low number of students obtaining the correct answer. Working showed that many of the candidates appreciated that the field needed to be resolved and were able to select the right trigonometrical function. However, by far the most common mistake was to ignore the turns and to simply calculate the flux through the coil. Whether this is a misconception or simply looking at the 'coil' on the diagram is unknown, however candidate should be reminded of the difference between turns and coils.
			Total	1	
3			D	1	<u>Examiner's Comments</u> The correct response is D. This question was correctly answered by the majority of candidates, although almost all the incorrect responses were C, presumably as candidates are aware that it is the e.m.f. that is induced but less familiar with Faraday's law in general.
			Total	1	
4			D	1	
			Total	1	
5			c	1	<u>Examiner's Comments</u> This was a well-answered question with most candidates correctly recalling that charge is conserved according to Kirchhoff's first law. A significant number of candidates distracted towards B; perhaps because of the unit of charge is the coulomb.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	1	
6			A	1	<u>Examiner's Comments</u> A majority of candidates were able to identify the correct response. The vast majority of incorrect responses were A, probably from the flux linkage at that point, rather than the rate of change of flux linkage.
			Total	1	
7			A	1	<u>Examiner's Comments</u> Just over half of the candidates correctly applied Fleming's Left Hand Rule to obtain the correct response. Candidates should be familiar with the terms 'into, down, out of, up' in relation to directions in this style of question.
			Total	1	
8			A	1	<u>Examiner's Comments</u> Most candidates who successfully answered this question showed working equating the magnetic force to the centripetal force to show that $mv = rBq$. Some had possibly learnt the formula $r = mv/Bq$ as this was also a common starting point.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
9			C	1	<u>Examiner's Comments</u> Most candidates appear to know that the force between the two wires would be attractive, and circled C or D as their options. A few used the diagram and the right hand rule to draw arrows to help with the direction. Both of these are good practice in this style of question. A little less than half of the candidates selected the correct response with the vast majority of incorrect responses being D.
			Total	1	
10			C	1	
			Total	1	
11			B	1	
			Total	1	
12			D	1	
			Total	1	
13			D	1	
			Total	1	
14			A	1	
			Total	1	
15			C	1	
			Total	1	
16			A	1	
			Total	1	
17			C	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
18			B	1	<u>Examiner's Comments</u> Many candidates were able to correctly complete this two-stage calculation. Most candidates showed clear working although several attempted to use a centripetal force equation. Many of the candidates who did not select the correct response did set up the equations correctly but seemed unable to recall the charge on the proton.
			Total	1	
19			B	1	<u>Examiner's Comments</u> This question was correctly answered by a good proportion of the candidates, as it was a relatively simple two step calculation. The vast majority of incorrect responses showed little useful working, showing a lack of knowledge of transformers.
			Total	1	
20			C	1	<u>Examiner's Comments</u> The majority of candidates were correctly able to identify Faraday's law from a list of physical laws. Many candidates knew that it was related to electromagnetism and so the majority of incorrect responses were A.
			Total	1	
21			B	1	
			Total	1	
22			C	1	
			Total	1	
23			D	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
24			C	1	
			Total	1	